

Kanea Chatbot

How to run it, open it, and make changes — a simple guide

1. What you need installed

- **Python 3.10+** (for the backend)
- **Node.js 18+** (for the frontend)
- A free **Groq API key** from console.groq.com — this powers the chatbot's replies. It already goes in `backend/.env` as `GROQ_API_KEY`.

If this is your first time running the project, install the dependencies once:

```
cd backend
pip install -r requirements.txt

cd ../frontend
npm install
```

2. Start the backend

The backend is the "brain" — it classifies messages, talks to the AI model, and stores everything in the database.

Open a terminal and run:

```
cd backend
python -m uvicorn main:app --reload --port 8000
```

Leave this terminal open. You'll know it worked when you see:

```
Uvicorn running on http://127.0.0.1:8000
```

3. Start the frontend

The frontend is the actual chat screen the customer sees.

Open a **second, separate** terminal (keep the backend one running) and run:

```
cd frontend
npm run dev
```

You'll see a message with a local URL, usually:

```
Local:   http://localhost:5173/
```

Important: the frontend needs the backend running first, otherwise the chat won't connect.

4. Open the chatbot

Open your browser and go to:

http://localhost:5173

You'll see three tabs:

- **Chat Support** — talk to Kanea, the AI assistant
- **Outage Status** — see live power outages
- **Report a Fault** — submit a fault report

5. Stopping everything

In each terminal, press `Ctrl + C`.

If a port is still stuck ("in use") afterward, on Windows you can free it with:

```
netstat -ano | findstr :8000
taskkill /PID <the number you see> /F
```

(Same for port `5173` if the frontend gets stuck.)

6. Admin dashboard (optional)

There's a separate admin site for viewing analytics, fault reports, and escalations. It's a different app in the `frontend-admin` folder:

```
cd frontend-admin
npm install      # first time only
npm run dev
```

It opens on its own port (check the terminal output) and asks for a password — that password is set in `backend/.env` as `ADMIN_PASSWORD`.

7. Where to go if you want to change something

Everything the chatbot does lives in a small number of files. Here's what each one is for, in plain terms:

Backend files (`backend/`)

File	What it does
<code>chatbot.py</code>	The actual "thinking" of the bot: figures out what the customer wants (billing? outage? emergency?), decides how to answer, and builds the message sent to the AI model. This is the file you'll edit most.
<code>knowledge_base.py</code>	Searches <code>ecg_knowledge.txt</code> for the most relevant facts to answer a question.
<code>ecg_knowledge.txt</code>	Plain text file of everything the bot "knows" about ECG — billing, safety, connections, etc. No coding needed to update this — just edit the text.
<code>database.py</code>	Saves and reads everything from the database: chat history, ratings, fault reports, outages.
<code>main.py</code>	Connects everything together as a web server — this is what the frontend actually talks to.
<code>ecg_scraper.py</code>	Automatically checks <code>ecg.com.gh</code> once an hour for real outage updates.
<code>eval/</code>	A testing tool that checks how accurate the bot is — not part of the live chatbot, just for measuring quality.

Frontend files (`frontend/src/`)

File	What it does
<code>components/ChatWindow.jsx</code>	The actual chat box — messages, typing indicator, quick-reply buttons, star ratings.
<code>App.jsx</code>	The page around the chat box — header, tabs, footer.
<code>components/FaultForm.jsx</code>	The "Report a Fault" form.
<code>components/OutageBoard.jsx</code>	The "Outage Status" list.
<code>tailwind.config.js</code>	All the brand colors (navy/red/yellow) in one place.
<code>config.js</code>	Where the frontend looks for the backend — only touch this if you change the backend's address/port.

8. Quick "I want to change X" cheat sheet

I want to...	Go to...
Change how the bot talks / its personality	<code>chatbot.py</code>
Teach the bot new facts / FAQs	<code>ecg_knowledge.txt</code>
Change when it hands off to a human	<code>chatbot.py</code> (danger/escalation keywords near the top)
Change the colors	<code>tailwind.config.js</code>
Change how chat messages look	<code>ChatWindow.jsx</code>
Change the fault report form	<code>FaultForm.jsx</code>
Add a new backend feature/endpoint	<code>main.py</code>
Change what's saved in the database	<code>database.py</code>